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Education

University of Florida—3.78/4.0

Aug. 2025 – June 2027

Masters in Artificial Intelligence Systems

Gainesville, Florida

MKSSS Cummins College of Engineering for Women Pune—3.5/4.0

Aug. 2021 – June 2025

Bachelor of Technology in Computer Engineering with Honors in Data Science and Machine Learning

Pune, Maharashtra

Relevant Coursework

- Machine Learning
- Data Structures
- Computer Vision
- Statistics
- Deep Learning
- Object Oriented Prog
- Database Management
- Business Intelligence

Internship

Center for Research Computing, University of Notre Dame

June 2024 – Dec. 2024

Research Intern

Notre Dame, Indiana

- Developed a Python-based intelligent agent using LLMs and Selenium that automated data acquisition from dark web marketplaces, scraping 200+ pages in 92 minutes including CAPTCHA solving and storing results in CSV.
- Evaluated 8 models (Claude Sonnet 3, 3.5, 3.7, 4, GPT-4, GPT-4.1, GPT-4o) for solving row-wise scrambled, column-wise scrambled and tile-rotate style scrambled image CAPTCHAs versions and observed highest accuracy of 93.33%.
- Applied computer vision techniques to optimize image inputs, improving model analysis reliability and speed.
- Presented experimental results at the Notre Dame university research symposium and co-authored a paper accepted in Lecture Notes in Artificial Intelligence (LNAI) series.

Projects

Co-Wingman: AI Conversational Support System | *Python, Tensorflow, SHAP, Gradio*

Dec. 2025

- Developed a real-time speech anxiety detection system using CNN-LSTM on MFCC features, achieving 96% recall with 1.2–1.5s inference latency on rolling 5-second audio windows.
- Integrated Whisper transcription, MiniLM keyword extraction, and SHAP explainability to provide interpretable, moment-level stress analysis from live conversations.
- Deployed the system using Docker Compose with Prometheus and Grafana monitoring that enabled real time latency tracking and confidence drift.

CruxCapture App | *Java, Android Studio, Firebase, OneAI API*

Dec. 2024

- Developed an Android app to summarize text, audio, and video using OneAI API, reducing content review time.
- Enabled user authentication and custom dashboards using Firebase, enhancing user engagement and data management.
- Implemented PDF generation and sharing via SMS and email using registered user IDs.
- Debugged backend API calls using Postman, ensuring communication between frontend and summarization services.

Efficient enumeration of active hidden servers on the Dark Web | *Python, Docker, Proton VPN, Linux*

Jan. 2023

- Created a Proof of Concept to detect active hidden (.onion) sites achieving recognition in Smart India Hackathon 2023.
- Used Docker containers with VPN tunneling for anonymization, ensuring privacy and compliance while on dark web.
- Leveraged OnionScan to scan dark web sites on www, identifying active hidden servers and validating their accessibility.

Research Papers

- **Interpretable Fake News Detection Using Neural Networks and LIME** published in ISD4SD 2025 Conference.
- **Multimodal Web Agents for Automated (Dark) Web Navigation** published in ICAART 2025 Conference.
- **USD to INR Exchange Rate Prediction: A Deep Learning Approach for Forecasting Currency Exchange Rates Using Different Techniques of LSTM** published in ICICT 2024 Conference.

Technical Skills

- **Programming Languages:** Python, Java, C, SQL, JavaScript, HTML, CSS
- **Machine Learning & AI:** Supervised & Unsupervised Learning, CNNs, RNNs, LSTM, Transformers, NLP, Speech Processing, Feature Engineering, Model Evaluation (F1, ROC-AUC), Explainable AI
- **Data Science & ML Frameworks:** Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Keras, OpenCV, NLTK, Librosa, Data Cleaning, EDA, Statistical Analysis, Data Visualization
- **Deployment & MLOps:** REST APIs, Flask, Streamlit, Gradio, Docker, Model Deployment, Monitoring (Prometheus, Grafana)
- **Developer Tools:** VS Code, GitHub, Eclipse, Postman, Android Studio
- **Cloud & Databases:** AWS (EC2, S3), GCP, MySQL
- **Visualization & Design:** Power BI, Tableau, Figma, Canva, Lucidchart